

NEURALYZING THE TRACE: SELECTIVE REPRESENTATION-LEVEL UNLEARNING WITH CONTRASTIVE SPARSE AUTOENCODERS

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ABSTRACT

Machine unlearning aims to remove targeted information while preserving a model’s other abilities. In realistic settings, such as privacy requests under the EU GDPR, the target may be narrow, for example information associated with a single person. Behavioral forgetting alone may be insufficient, motivating interventions directly on internal representations. However, standard mechanistic-interpretability extractors are poorly selective for such targets. We identify an *energy bias* in reconstruction-based extraction, which favors dominant background structure over low-energy target-specific components. We introduce SCALPEL, a contrastive sparse autoencoder designed to learn more selective forget features. We show theoretically that contrastive training promotes target-selective features and that our selection score controls expected background knowledge perturbation. We validate SCALPEL experimentally on TOFU across Qwen, Llama, and Gemma, where it substantially improves over NMF and standard SAE interventions and is competitive with Gradient Difference and RMU, bridging mechanistic interpretability and fine-grained unlearning.

1 INTRODUCTION

Once a large language model (LLM) has learned a piece of information, it is hard to take back. Yet a private fact, a copyrighted passage, or a narrow sensitive association may need to be removed without retraining from scratch. This is the goal of machine unlearning, and it matters directly for privacy, model governance, and legal requirements such as the right to erasure (“right to be forgotten”) established in Article 17 of the EU General Data Protection Regulation (GDPR) (European Parliament and Council of the European Union, 2016).

Most existing methods treat unlearning as an optimization problem on the model weights: gradient-difference methods (Bu et al., 2024), retain-only finetuning, representation steering such as RMU (Huu-Tien et al., 2024), or in-context unlearning (Pawelczyk et al., 2023). These methods can make the model stop answering forget-set queries. But they leave a basic question open: did the model remove the knowledge, or did it only learn not to express it?

If the internal representation survives, the knowledge can resurface under paraphrases, indirect prompts, jailbreak-style queries, or a small amount of relearning. Recent work on parametric knowledge traces shows that some unlearning methods change behavior while leaving much of the underlying trace intact (Hong et al., 2025). This motivates a more direct approach: identify target-associated representations inside the model and intervene on them explicitly.

Mechanistic interpretability provides natural tools for this. Transformer computation can be read through the residual stream, where information accumulates across layers (Elhage et al., 2021), and concept-extraction methods such as PCA, NMF, and sparse autoencoders (SAEs) decompose activations into interpretable directions (Fel et al., 2023; Jourdan et al., 2023; Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024). But these tools work best when the target is a broad concept already well separated in activation space (Farrell et al., 2024; Patil, 2024), whereas realistic requests are far more precise: forgetting one author is not removing a whole topic. In this regime

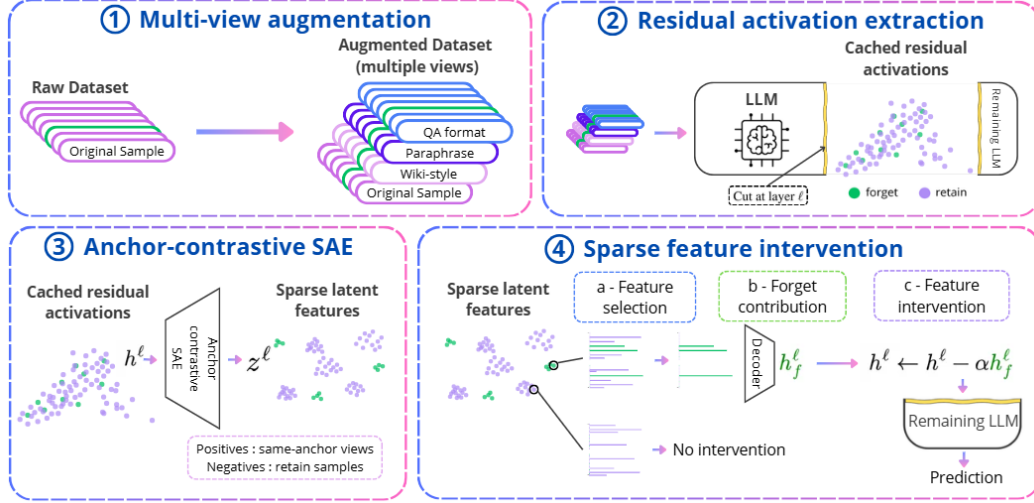


Figure 1: Overview of the SCALPEL pipeline. ① We generate multiple semantic views of each forget anchor, then ② cache residual-stream activations at layer ℓ . ③ An anchor-contrastive SAE learns sparse features using same-anchor views as positives and **retain samples** (purple) as negatives, so that **forget samples** (green) become isolated from retain information in the sparse feature space. ④ We select forget-specific features, decode their contribution h_f^ℓ , and subtract it from the residual stream before the remaining layers produce the prediction. The same selected feature set is applied to every input, while subtraction occurs only when those features activate. The intervention therefore acts as an input-dependent internal filter intended to suppress forget-related components while limiting its effect on retain activations.

we find that standard PCA- and SAE-based interventions are poorly selective, and the features they remove damage retain knowledge almost as much as forget knowledge.

One important cause lies in the training objective. Indeed, Reconstruction-based extractors are rewarded in proportion to the *energy* (dispersion) of what they capture, and a specific forget target is a low-energy (low dispersion) component hidden inside a much larger background (high dispersion) (see Section 3). The extractor is therefore more strongly rewarded for modeling the background than for isolating the target.

We address this with SCALPEL, a contrastive sparse autoencoder built on a simple hypothesis: specific knowledge is easier to isolate when the extractor is trained to focus on what stays stable across many views of the same target. We augment each forget anchor into paraphrases and rewritten variants, treat views of the same anchor as positives, and draw negatives from retain data rather than from other forget examples. This avoids pushing related forget examples apart and pulls target-specific structure away from shared language and task features. Once trained, the SAE provides a fixed set of forget-associated sparse features. The same set is used for every input, while each feature is subtracted only when activated by the encoder. This produces an explicit and input-dependent internal filter with tunable strength (Figure 1).

The goal of this paper is not to solve machine unlearning in full or to provide certified deletion from the model parameters. Instead, we take a first step toward connecting mechanistic interpretability with targeted unlearning by studying whether internal features can be extracted with enough selectivity to support narrow forget requests.

Contributions. We study this question in a controlled author-level setting on TOFU and show that adapting the *feature-learning objective* can produce selective internal features that support competitive representation-level unlearning while remaining explicit and inspectable. Our contributions are:

- We identify and formalize a **selectivity bottleneck**: reconstruction-trained extractors are energy-weighted and may therefore underrepresent low-energy forget targets (Section 3).
- We introduce **SCALPEL**, an anchor-contrastive sparse autoencoder that learns features stable across augmented views of a target and weak on background data, together with a **norm-aware selection criterion** accounting for both encoder activation and decoder-direction magnitude (Sections 4–4.5).
- We provide a **theoretical account** of the method: reconstruction objectives are energy-weighted, adding contrastive learning favors more target-selective sparse representations, and high-score features admit a bound on expected background perturbation (Section 5).
- We **validate the approach empirically** on TOFU across three model families, showing improved feature selectivity and competitive unlearning performance against both mechanistic-interpretability and optimization-based baselines (Section 7).

2 RELATED WORK

General approaches to LLM unlearning. Unlearning removes the influence of a forget set $\mathcal{D}_f$ while preserving behavior on a retain set $\mathcal{D}_r$; an auxiliary set $\mathcal{D}_w$ (e.g. Wiki data) is often used to check general knowledge (Maini et al., 2024; Shi et al., 2024; Dorna et al., 2025). A first family updates the weights with gradient-based objectives. Gradient Ascent raises the loss on forget examples but quickly damages the model (Maini et al., 2024); Gradient Difference adds descent on $\mathcal{D}_r$ to limit the damage, with normalized variants (Maini et al., 2024; Bu et al., 2024); Negative Preference Optimization replaces the unstable ascent term with a smoother preference-based loss (Zhang et al., 2024); other approaches finetune the model toward generic or incorrect answers (Eldan & Russinovich, 2023). Unlearning can also avoid weight changes: in-context unlearning conditions the model at inference time (Pawelczyk et al., 2023), which changes behavior but not the stored representation. Model editing such as ROME updates parameters tied to one factual association (Meng et al., 2022), though editing replaces an association rather than removing an influence.

A particularly relevant baseline is Representation Misdirection for Unlearning (RMU) (Li et al., 2024; Huu-Tien et al., 2024), which pushes forget activations at a chosen layer toward a random direction while pinning retain activations to those of the frozen original model. RMU shares our core principle act on internal representations, not just outputs but never identifies an interpretable representation of the target: it moves the whole forget activation and modifies the weights. SCALPEL instead extracts a small set of sparse target-specific features and intervenes only along their decoded directions. Comparing against RMU therefore tests whether an interpretable, selective intervention can match a stronger but opaque one.

Behavioral evaluation alone cannot establish that knowledge is gone: an internal trace may remain recoverable through other prompts or light finetuning (Hong et al., 2025; Yuan et al., 2024). Representation-level methods do not guarantee parametric deletion either, but they make the mechanism explicit by causally linking behavioral change to an identified internal intervention.

Mechanistic-interpretability-based unlearning. In transformers, a natural intervention point is the residual stream, through which information flows between layers (Elhage et al., 2021; Dar et al., 2022; Geva et al., 2022; 2023). Concept-decomposition methods express activations through a small set of meaningful directions: PCA extracts high-variance orthogonal directions and NMF factorizes non-negative activations (Jolliffe, 2002; Lee & Seung, 1999); we use NMF as a classical baseline whose components can be scored on forget/retain data and used as intervention directions. Sparse autoencoders offer a more flexible decomposition and have become a standard tool for interpretable features (Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Bussmann et al., 2024; Rajamanoharan et al., 2024). Given an activation h , an SAE computes a sparse code $z(h) = E_\phi(h)$ and reconstructs $\hat{h} = b_D + \sum_{q=1}^Q z_q(h) d_q$, where b_D is the decoder bias and d_q the decoder direction of feature q ; the linear decoder associates each feature with an explicit direction in residual space, making feature-level interventions decomposable and inspectable.

A standard SAE unlearning pipeline trains the SAE, identifies forget-related features from activation statistics, and modifies them at inference. A direct intervention writes $\tilde{h} = h - \alpha \sum_{q \in S_f} z_q(h) d_q$ for a selected set S_f and strength α , with clamping or negative scaling as variants (Farrell et al.,

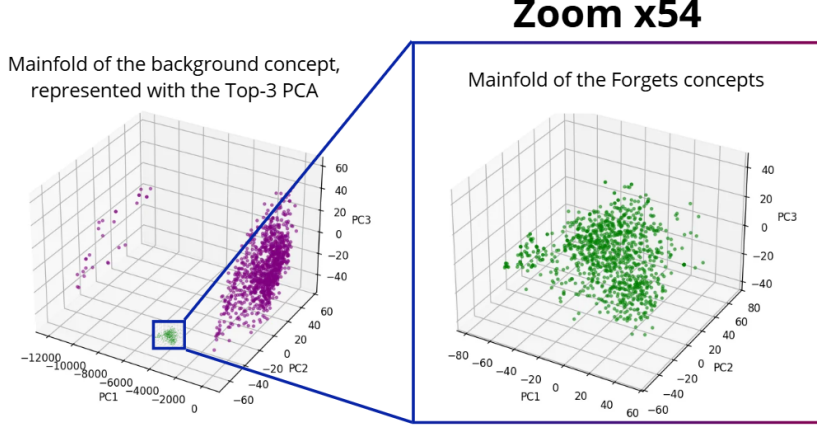


Figure 2: Energy comparison between dominant background structure and background-orthogonal forget residuals at layer -4 of Qwen. Left: background activations projected onto the first three principal components learned from retain and Wiki activations. Right: forget activations after removing these directions, shown in a second PCA basis fitted to the residuals. Since the panels use different bases, they should be compared through their scale and explained energy, not their coordinates. The background top-3 PCA energy is $\sim 329\times$ larger.

2024). An alternative is to edit the sparse code and replace h by the full SAE reconstruction, as in LLM Neurosurgeon (Zhou et al., 2025). That choice, however, additionally injects the SAE’s reconstruction error, $\tilde{h} - h = -\sum_{q \in S_f} z_q(h) d_q + (b_D + Dz(h) - h)$, which in our experiments degraded retain and Wiki performance even before any feature was removed. We therefore always intervene on the original activation. SAE features can also constrain weight updates through decoder-direction subspaces (Wang et al., 2025); this suits broad, globally separable topics, whereas our targets overlap heavily with retain data and do not occupy a clean linear subspace. Existing MI-based unlearning methods generally reuse features learned by generic reconstruction objectives; we instead study how to train the extractor itself for the selectivity required by narrow forget targets.

The main limitation of these pipelines is not finding features that activate on forget examples, but finding features that activate *specifically* on the target while remaining weak on shared background structure. SCALPEL addresses this selectivity bottleneck by adapting the feature-learning objective itself. The next section explains why reconstruction objectives are poorly aligned with this requirement.

3 ENERGY BIAS IN RECONSTRUCTION-BASED EXTRACTION

One important source of the selectivity bottleneck is an objective mismatch: classical extractors optimize reconstruction, whereas targeted unlearning requires target specificity. Following the additive concept-manifold view of Bhalla et al. (2026), suppose an activation decomposes as $h = \sum_{j=1}^J m_j$, with $m_j \in \mathcal{M}_j$ (Mainfold j), and define its energy as $E_j = \mathbb{E}[\|m_j\|_2^2]$. Under centered, mutually uncorrelated concept components, the reconstruction loss of an extractor Θ decomposes as

$$\mathcal{L}_{\text{rec}}(\Theta) = \sum_{j=1}^J E_j (1 - C_j(\Theta)),$$

where $C_j(\Theta) \in [0, 1]$ is the fraction of concept j ’s energy captured by Θ . Improving its capture by ΔC_j yields a reconstruction gain of $E_j \Delta C_j$, so the objective assigns a larger marginal gain to a background concept $\mathcal{M}_b$ than to a forget concept $\mathcal{M}_f$ whenever $E_b \Delta C_b > E_f \Delta C_f$ (Appendix A and Figure 2). This is a stylized account of the incentive induced by reconstruction, not an impossibility result for trained SAEs: a forget concept may remain important to the model while contributing little variance. Consequently, PCA, NMF, and reconstruction-only SAEs are more strongly rewarded

for modeling high-energy structure, and a low-energy but precise target may be blended into broader features.

We examine this on residual-stream activations from layer -4 of Qwen. We fit PCA on the background activations $H_{\text{bg}} = H_r \cup H_w$, remove their top-three directions from the forget activations, and fit a second PCA on the residuals (Appendix A). The corresponding top-three energies are $\hat{E}_{\text{bg}}^{(3)} = 3,787,695$ and $\hat{E}_f^{\perp \text{bg},(3)} = 11,517$, a factor of about 329. Although removing the background directions means that the residuals are not a pure forget manifold, the gap is far larger than under a random split of the same data. This is consistent with the proposed mechanism: target-specific structure can contribute far less reconstruction energy than the background from which it must be separated, motivating an objective that explicitly rewards *selectivity*.

4 SCALPEL: SPARSE CONTRASTIVE AUTOENCODERS FOR PRECISE REPRESENTATION-LEVEL UNLEARNING

SCALPEL defines fine-grained forget targets, constructs multiple semantic views, learns target-selective sparse features from cached residual activations, and filters their decoded contribution at inference (Figure 1).

4.1 TARGET GRANULARITY AND MULTI-VIEW CONSTRUCTION

Targeted unlearning first needs a precise definition of a forget concept. Let $\mathcal{D}_f, \mathcal{D}_r, \mathcal{D}_w$ be the forget, retain, and Wiki datasets and $\mathcal{D}_{\text{bg}} = \mathcal{D}_r \cup \mathcal{D}_w$. In TOFU, forget and retain authors come from the same generation process and the split is random, not semantic: they share writing style, factual structure, vocabulary, and QA format, and their activations overlap heavily (Appendix D, Figure 5). Treating the whole forget set as one concept would push the extractor toward features of the common TOFU format rather than the knowledge to remove. We instead define one concept per forget author, $G_a = \{x \in \mathcal{D}_f : \text{author}(x) = a\}$, which matches the right-to-be-forgotten setting and gives several related samples per target (sample-level granularity is discussed in Appendix I).

For each target G_a we build a set of augmented views $\mathcal{V}_a = \{x_{a,1}, \dots, x_{a,m_a}\}$ using paraphrasing, prompt-style rewriting with Qwen2.5-7B-Instruct, and backtranslation (Sennrich et al., 2016; NLLB Team et al., 2022), in the spirit of view-based contrastive learning (Chen et al., 2020). Before augmentation we extract the key facts of the target (names, dates, titles, relations) and constrain them to remain unchanged, so the surface form varies while the defining target facts are retained. The same recipe extends to document-level datasets such as MUSE by treating each chunk as a target; we leave that evaluation to future work.

4.2 RESIDUAL-STREAM ACTIVATION EXTRACTION

We run the finetuned model and cache residual-stream activations $h_t^\ell(x) \in \mathbb{R}^d$ at a fixed layer ℓ , in practice the fourth layer before the output: deep enough to carry high-level factual information, but before the final layers specialize for next-token prediction (Yu et al., 2024; Mabrok, 2026). We keep this layer fixed across all representation-based methods. We do not claim that it is optimal, and a systematic study of layer selection is left for future work. For a dataset $\mathcal{D}_s$ and retained token positions $\mathcal{I}(x)$, we write $\mathcal{H}_s = \{h_t^\ell(x) : x \in \mathcal{D}_s, t \in \mathcal{I}(x)\}$ for $s \in \{f, r, w\}$, $\mathcal{H}_{\text{bg}} = \mathcal{H}_r \cup \mathcal{H}_w$, and $\mathcal{H}_{G_a}$ for the tokens of target G_a . Each activation dimension is standardized with statistics from the training cache, $\bar{h} = (h - \mu_H) \oslash \sigma_H$; both the SAE input and its reconstruction live in this standardized space, which we found important for robustness across model families.

4.3 CONTRASTIVE SPARSE AUTOENCODER

Sparse decomposition. All variants share the same base. The encoder maps a standardized activation to a non-negative sparse code $z(\bar{h}) = E_\phi(\bar{h}) \in \mathbb{R}_+^Q$, and a linear decoder reconstructs $\hat{\bar{h}} = b_D + \sum_{q=1}^Q z_q(\bar{h}) d_q$. Training combines the usual reconstruction loss $\mathcal{L}_{\text{rec}} = \mathbb{E}_{\bar{h}}[\|\bar{h} - \hat{\bar{h}}\|_2^2]$ over forget, retain, and Wiki activations with an ℓ_1 penalty $\mathcal{L}_{\text{sparse}} = \mathbb{E}_{\bar{h}}[\|z(\bar{h})\|_1]$.

The raw coefficient z_q alone does not measure a feature’s decoded contribution and is sensitive to reciprocal rescaling of the encoder and decoder. We therefore use the scale-invariant norm-aware activation $\tilde{z}_q(\bar{h}) = z_q(\bar{h}) \|d_q\|_2 \geq 0$, and represent token $i = (x, t)$ by $r_i = (\tilde{z}_1(\bar{h}_i), \dots, \tilde{z}_Q(\bar{h}_i)) \in \mathbb{R}_+^Q$. The contrastive objective operates directly on these token-level codes through the cosine similarity $\text{sim}(i, j) = \langle r_i, r_j \rangle / (\|r_i\|_2 \|r_j\|_2)$; no token averaging is done before the similarity (sample-level aggregation is used only later, for feature scoring).

Token-level multi-positive InfoNCE. All cached activations are flattened into one pool of tokens, each inheriting the concept identifier and split of its source sample; the pool is shuffled and cut into minibatches each epoch, independently of sample boundaries. For an anchor token i , positives $\mathcal{P}(i)$ are the other in-batch tokens sharing its concept identifier (another token of the same sample, or a token of another view of the same target), and negatives $\mathcal{N}(i)$ are in-batch tokens with a different concept identifier that satisfy the split-specific contrastive mask; depending on the variant, negatives may include retain tokens or Wiki tokens. Anchors with an empty positive or negative set are dropped. With temperature $\tau > 0$ and $\kappa(i, j) = \exp(\text{sim}(i, j)/\tau)$, define the positive and negative masses $A_i^+ = \sum_{j \in \mathcal{P}(i)} \kappa(i, j)$ and $A_i^- = \sum_{j \in \mathcal{N}(i)} \kappa(i, j)$, and the multi-positive InfoNCE loss (van den Oord et al., 2018; Khosla et al., 2020)

$$\ell_{\text{NCE}}(i) = -\log \frac{A_i^+}{A_i^+ + A_i^-}, \quad \mathcal{L}_{\text{NCE}} = \frac{1}{|\mathcal{I}_{\text{valid}}|} \sum_{i \in \mathcal{I}_{\text{valid}}} \ell_{\text{NCE}}(i).$$

This pulls together token codes of the same target and pushes them away from other concepts and from background data. Crucially, negatives come from retain data rather than from other forget examples, so related forget samples are never pushed apart.

4.4 SCALPEL

SCALPEL. Forget and background samples share a lot of structure, and forcing one latent code to serve both reconstruction and separation can be restrictive. Inspired by contrastive SAEs of (Poupart, 2024), we split the code into two blocks produced by two encoders, $z(\bar{h}) = [z^{\text{com}}(\bar{h}), z^{\text{diff}}(\bar{h})]$, decoded jointly. The common block captures structure shared across samples ($\mathcal{L}_{\text{com}}$); the target-specific block contrasts views of the same forget target against retain and Wiki examples ($\mathcal{L}_{\text{diff}}$). The objective is $\mathcal{L}_{\text{split}} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{sparse}} + \beta_{\text{com}} \mathcal{L}_{\text{com}} + \beta_{\text{diff}} \mathcal{L}_{\text{diff}}$, and only features of the target-specific block are eligible for intervention.

4.5 FEATURE SELECTION AND RESIDUAL-STREAM INTERVENTION

After training, we score each feature by how active it is on one target relative to the background. With $\mu_q(G_a) = \mathbb{E}_{\bar{h} \in \mathcal{H}_{G_a}} [\tilde{z}_q(\bar{h})]$, and $\mu_q(\text{bg}) = \mathbb{E}_{\bar{h} \in \mathcal{H}_{\text{bg}}} [\tilde{z}_q(\bar{h})]$, the target-selectivity score is

$$s_q(G_a) = \frac{\mu_q(G_a)}{\mu_q(\text{bg}) + \varepsilon}. \quad (1)$$

For each target, we select $S_{G_a} = \{q : s_q(G_a) \geq s_{\min}\}$, with a top-feature fallback if the set is empty. For a forget request containing several targets, the fixed global intervention set is $S_f = \bigcup_{a \in \mathcal{A}_f} S_{G_a}$, where shared features are included only once.

The same set S_f is applied to every input, without access to author identity or forget/retain labels; its effect stays input-dependent, since a selected feature is subtracted only when the encoder activates it. As the SAE operates in standardized space, we compute $\bar{h}^\ell = (h^\ell - \mu_H) \odot \sigma_H$ and map the decoded contribution back:

$$\tilde{h}^\ell = h^\ell - \alpha \sum_{q \in S_f} z_q(\bar{h}^\ell) (\sigma_H \odot d_q). \quad (2)$$

Modifying the original activation rather than replacing it with the full reconstruction avoids injecting reconstruction error, and each component remains a sparse activation times a decoder direction.

5 THEORETICAL ANALYSIS

We formalize two properties of SCALPEL: adding contrastive learning to reconstruction and sparsity favors more target-selective features, and selecting such features limits their expected effect on background activations. Complete assumptions and proofs are given in Appendices B and C.

Contrastive learning promotes target-selective features. Write $\mathcal{L}_{\text{SAE}} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{sparse}}$ and $\mathcal{L}_{\text{SCALPEL}} = \mathcal{L}_{\text{SAE}} + \beta \mathcal{L}_{\text{NCE}}$. Reconstruction maintains an informative representation and sparsity concentrates it on few coordinates, but neither explicitly rewards target-background separation. The contrastive term provides this missing signal.

Let $\tilde{z}_q(h) = z_q(h)/\|d_q\|_2$ be the norm-aware code used by InfoNCE and $u_i = \tilde{z}(h_i)/\|\tilde{z}(h_i)\|_2$ its normalized representation.

Proposition 5.1 (Contrastive regularization favors target selectivity). *Assume non-negative codes, controlled decoder scales, and that a sparse target-background separation is compatible with comparable reconstruction and sparsity cost. Among such representations, adding $\mathcal{L}_{\text{NCE}}$ favors $\langle u_i, u_j \rangle \rightarrow 1$ for same-target positives and $\langle u_i, u_j \rangle \rightarrow 0$ for background negatives. More precisely, if the contrastive positive-to-negative mass ratio is within a factor $1 + \delta$ of its optimum, then*

$$\frac{1}{|N(i)|} \sum_{j \in N(i)} \langle u_i, u_j \rangle \leq \tau \delta \quad \text{and} \quad \frac{1}{|P(i)|} \sum_{j \in P(i)} \|u_i - u_j\|_2^2 \leq \frac{2\delta}{(1 + \delta)(1 - e^{-1/\tau})}$$

Thus, as the contrastive objective improves, views of the same target share increasingly similar sparse coordinates while their overlap with background codes decreases. Combined with sparsity, this favors features that are active on a target and weak on the background, precisely the structure measured by the selectivity score $s_q(G_a)$ in Eq. equation 1. Different forget targets may still share features when they are not contrasted against each other.

Selectivity controls background perturbation. In standardized activation space, let $\delta_{G_a}(\bar{h}) = \alpha \sum_{q \in S_{G_a}} z_q(\bar{h}) d_q$.

Proposition 5.2 (Selectivity controls background perturbation). *If every selected feature satisfies $s_q(G_a) \geq s_{\min}$, then*

$$\mathbb{E}_{\bar{h} \sim \bar{\mathcal{H}}_{\text{bg}}} [\|\delta_{G_a}(\bar{h})\|_2] \leq \alpha \sum_{q \in S_{G_a}} \mu_q(\text{bg}) \leq \frac{\alpha}{s_{\min}} \sum_{q \in S_{G_a}} \mu_q(G_a)$$

The bound controls expected representation-space perturbation, not downstream behavior or worst-case effects. It also ignores the orientation of d_q , so similarly scored features may have different causal effects, as examined in Section 7.3.

Together, the results explain the intended mechanism: reconstruction preserves information, sparsity concentrates it, contrastive learning organizes the sparse code according to the target concepts, and the resulting selectivity limits the expected background footprint of the intervention.

6 EXPERIMENTS

We run one main benchmark and two diagnostics. The benchmark compares representation-level unlearning performance against classical, mechanistic-interpretability, and optimization-based baselines across three model families. The diagnostics test whether contrastive training produces more target-selective sparse-code geometry and whether the resulting selectivity score predicts intervention quality.

Models and reference checkpoints. We use Qwen2.5-1.5B-Instruct, Llama-3.2-1B-Instruct, and Gemma-2-2B-it (Yang et al., 2024; Grattafiori et al., 2024; Gemma Team, 2024). For each model family and seed, we produce two references: a *full* model θ_{full} finetuned on $\mathcal{D}_f \cup \mathcal{D}_r$, which is the starting point of every unlearning method, and a *retain-only* model θ_{ret} finetuned on $\mathcal{D}_r$ alone, which approximates the desired endpoint. Within each seed, all methods start from the same full checkpoint. Augmented views are generated separately with Qwen2.5-7B-Instruct.

Dataset and augmentation. We evaluate on TOFU (Maini et al., 2024), using Wiki data $\mathcal{D}_w$ to measure general-knowledge preservation. For each forget author, we construct augmented views using 16 templates, including question–answer reformulations, biographies, Wiki-style passages, paraphrases, multi-hop questions, and adversarially phrased instructions (Zou et al., 2023; Dahal et al., 2026). We then backtranslate them through French, Spanish, and German with NLLB-200 (Sennrich et al., 2016; NLLB Team et al., 2022). The transformations vary wording and format while preserving the names, dates, titles, and relations defining the target.

Compared methods. We compare **NMF** (Lee & Seung, 1999); a **standard SAE** trained with reconstruction and sparsity only (Cunningham et al., 2023; Farrell et al., 2024); **contrastive SAEs** with a single shared code, using one- or two-layer encoders; **SCALPEL**; **Gradient Difference** (Bu et al., 2024); and **RMU** (Huu-Tien et al., 2024). All representation-based methods use layer $L - 4$, the same token positions, and the same selection and intervention procedure whenever the decomposition permits it, so the comparison isolates the effect of the learned representation.

Protocol. Each method is tuned independently for every model family using Optuna (Akiba et al., 2019), with 25 trials per model–method pair and the search spaces reported in Appendix E. The selected configuration is frozen and rerun over five independent end-to-end seeds. Each seed repeats reference-model finetuning, method training, feature selection, and evaluation with OpenUnlearning (Dorna et al., 2025).

Metrics and trade-off curves. Our main trade-off compares TOFU model utility with forget-set ground-truth answer probability (Maini et al., 2024). We additionally report Q–A probability and ROUGE on forget, retain, and Wiki data, together with forget quality, truth ratio, privacy leakage, extraction strength, and paraphrase-based evaluation (Lin, 2004; Maini et al., 2024; Shi et al., 2024; Carlini et al., 2021; Dorna et al., 2025). For methods exposing an intervention strength α , we trace the forget–retain trade-off using adaptive bisection in $\log \alpha$ and interpolate the five seed-specific curves onto a common grid (Appendix H). Complete multi-metric trade-offs are reported in Appendix F.2.

Diagnostics. The *contrastive-geometry diagnostic* (Section 7.2) compares the standard SAE, contrastive SAE, and SCALPEL-Split across the five Qwen benchmark seeds. On held-out activations, we normalize the norm-aware codes as $u = \tilde{z}/\|\tilde{z}\|_2$. For each forget target G_a , we measure the within-target similarity $C_{\text{pos}}(G_a) = \mathbb{E}_{i,j \in G_a}[\langle u_i, u_j \rangle]$ and the target–background similarity $C_{\text{bg}}(G_a) = \mathbb{E}_{i \in G_a, j \in \text{bg}}[\langle u_i, u_j \rangle]$. Higher C_{pos} and lower C_{bg} indicate a more target-selective representation. We then compare feature selectivity across methods to test whether this geometric separation translates into more selective sparse features.

The *score-quality diagnostic* (Section 7.3) considers the top-5 feature–target pairs for each of ten authors, giving 50 selections. Each pair is evaluated alone by sweeping α and tracing its $(P_{\text{forget}}, P_{\text{retain}})$ trajectory, which we compare across selectivity-score quantile bins.

7 RESULTS

7.1 MAIN UNLEARNING BENCHMARK

Figure 3 shows the main benchmark on Qwen; Llama and Gemma behave consistently and are reported in Appendix F.1. Three patterns stand out. First, NMF and the standard SAE are poorly selective: reductions in forget probability come with substantial loss in model utility, as predicted by the objective mismatch of Section 3 their features fire on forget examples but also capture structure shared with retain and Wiki data.

Second, contrastive feature learning substantially improves the trade-off. The plain contrastive SAEs already outperform the reconstruction-only SAE, while the main SCALPEL variants lie on or close to the Pareto front over the practically relevant range. This supports our central claim that adapting the extraction objective to target selectivity produces more effective representation-level interventions.

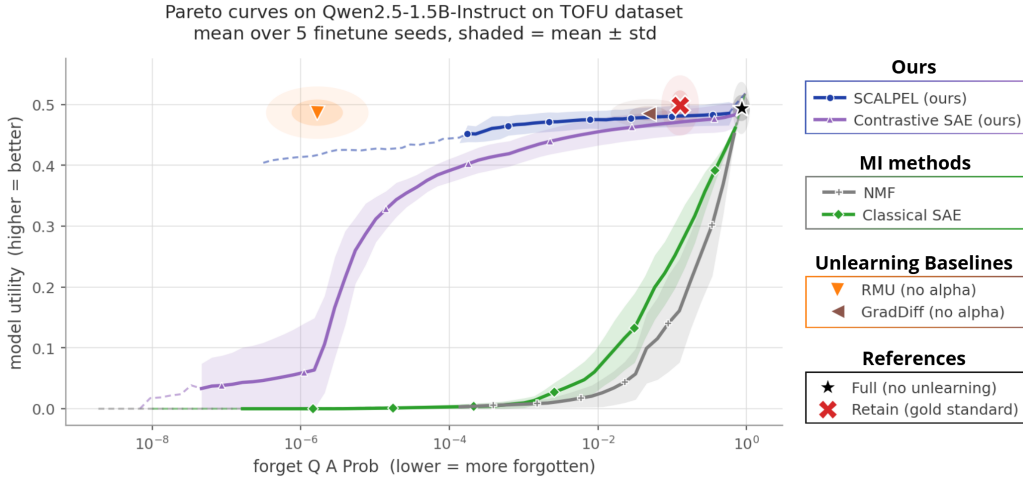


Figure 3: Unlearning trade-off on TOFU forget10 for Qwen2.5-1.5B-Instruct, after method-specific hyperparameter optimization and aggregation over five independent end-to-end seeds. Each curve varies the intervention strength: lower forget probability indicates stronger forgetting, while higher model utility indicates better preservation. Results for Llama and Gemma are reported in Appendix F.1 are similar. For more metrics look at Appendix F.2.

Third, RMU and Gradient Difference remain strong baselines, but SCALPEL achieves competitive trade-offs while keeping the base model fixed and expressing the intervention through explicit sparse feature activations and decoder directions. Corresponding experiments on Llama and Gemma are provided in Appendix F.1.

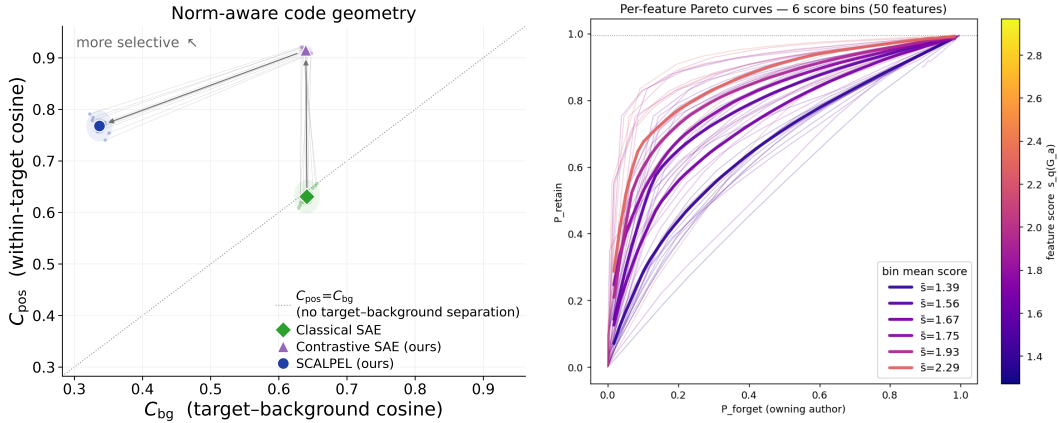
7.2 CONTRASTIVE LEARNING INCREASES FEATURE SELECTIVITY

Figure 4a tests the geometric prediction of Proposition 5.1. The reconstruction-only SAE lies close to $C_{\text{pos}} = C_{\text{bg}}$, with a separation gap $\Delta C = C_{\text{pos}} - C_{\text{bg}} \approx -0.01$, indicating little target-background separation. Adding the contrastive objective increases within-target similarity and raises the gap to $\Delta C \approx 0.27$. SCALPEL-Split further reduces target-background similarity, reaching $\Delta C \approx 0.43$. The representation therefore moves progressively toward the high- C_{pos} , low- C_{bg} regime favored by the sparse contrastive objective.

This geometric separation is also reflected at the feature level. Appendix G compares the top-5 selectivity scores of the same models across authors and seeds. Mean top-5 selectivity increases from approximately 1.9 for the reconstruction-only SAE to 4.0 for the contrastive SAE, and to 24.5 for SCALPEL-Split. Together, these results support the proposed role of contrastive learning: reconstruction and sparsity produce a compact representation, while the contrastive objective organizes its features around the target concepts and away from background structure.

7.3 THE SELECTIVITY SCORE PREDICTS UNLEARNING QUALITY

Figure 4b tests whether Eq. equation 1 predicts the quality of individual interventions. Each author-feature pair is evaluated in isolation while varying its intervention strength. In aggregate, higher-score bins preserve more retain probability at fixed forget probability, or achieve stronger forgetting at comparable retain probability. The selectivity score is therefore associated not only with feature selection but also with downstream intervention quality. Together with the contrastive-geometry diagnostic, this connects the learned representation to feature selectivity and, in turn, to intervention quality. The ordering remains imperfect at the level of individual features: the score captures activation magnitude and target-background selectivity, but ignores the orientation of d_q in the local activation geometry. A geometry-aware score would require substantially more expensive local manifold estimation (Appendix I).



(a) Norm-aware sparse-code geometry across the five Qwen benchmark seeds. For each method, C_{pos} measures within-target cosine similarity and C_{bg} target-background similarity on held-out activations. More selective representations lie toward the top-left. Adding contrastive learning increases within-target alignment, while SCALPEL-Split further reduces background overlap.

(b) Single-feature unlearning curves grouped by selectivity-score quantile. Thin curves represent individual author-feature pairs; thick curves show the average within each bin. Higher-score features generally provide better forget-retain trade-offs.

Figure 4: Diagnostics linking representation geometry, feature selectivity, and intervention quality for SCALPEL. Left: contrastive training increases within-target alignment and reduces target-background overlap in the norm-aware sparse code. Right: features with higher selectivity scores generally achieve better single-feature forget-retain trade-offs.

8 CONCLUSION AND LIMITATIONS

We proposed a first approach for connecting mechanistic interpretability with fine-grained unlearning of specific information. Our study shows that reconstruction-based concept extractors can lack the selectivity required to isolate a narrow forget target from the background knowledge that should be preserved. We identify one mechanism behind this behavior: reconstruction rewards captured energy, while target-specific information may correspond to a low-energy component entangled with much larger background structure; at the studied layer, we measure a $\sim 329\times$ energy gap. SCALPEL addresses this problem by adapting the feature-learning objective rather than the intervention pipeline, contrasting multiple views of each target against background data to learn sparse, target-selective features. Across Qwen, Llama, and Gemma on TOFU, SCALPEL substantially improves over NMF and reconstruction-only SAE interventions and is competitive with RMU and Gradient Difference. Our theoretical analysis and diagnostics further show how contrastive learning promotes target-selective representations and how feature selectivity relates to intervention quality.

Limitations. Our evaluation covers TOFU and three relatively small models; larger models, less controlled datasets, and document-, fact-, or sample-level targets remain untested. The theoretical analysis is local, and the perturbation bound controls representation-space magnitude rather than model behavior. Most importantly, SCALPEL does not erase the parametric trace: it applies a persistent internal filter to selected feature contributions whenever they activate, so our results establish selective behavioral and representation-level unlearning under intervention, not certified deletion. Using the extracted features to guide permanent parameter updates while preserving their selectivity is the natural next step. A fuller discussion is in Appendix I.

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A ENERGY BIAS OF RECONSTRUCTION-BASED EXTRACTION

This appendix gives the formal statement behind Section 3: reconstruction-only objectives reward an extractor in proportion to the *energy* of what it captures, so a low-energy forget concept is systematically deprioritized relative to high-energy background structure. The result is a motivation, not a theorem about trained SAEs: it isolates the incentive created by the reconstruction loss under an idealized concept decomposition, in the spirit of the additive concept-manifold view of Bhalla et al. (2026). We then describe in full the PCA experiment that measures the corresponding energy gap on real activations (Figure 2).

A.1 SETTING

Let $h \in \mathbb{R}^d$ be a random residual-stream activation, and suppose it decomposes into concept-manifold components

$$h = \sum_{j=1}^J m_j, \quad m_j \in \mathcal{M}_j, \quad (3)$$

where $\mathcal{M}_j$ denotes the j -th concept manifold. Define the energy of concept j as

$$E_j = \mathbb{E}[\|m_j\|_2^2] > 0. \quad (4)$$

Let Θ denote an extractor (for instance an SAE), producing a reconstruction $\hat{h}^\Theta$ of h , with reconstruction loss $\mathcal{L}_{\text{rec}}(\Theta) = \mathbb{E}[\|h - \hat{h}^\Theta\|_2^2]$.

Assumption A.1 (Component orthogonality). *The concept components are centered and mutually uncorrelated:*

$$\mathbb{E}[m_j] = 0 \quad \text{for all } j, \quad \mathbb{E}[\langle m_j, m_k \rangle] = 0 \quad \text{for } j \neq k.$$

Under Assumption A.1, the total activation energy splits exactly across concepts, $\mathbb{E}[\|h\|_2^2] = \sum_{j=1}^J E_j$, which is what makes E_j interpretable as concept j 's share of the signal.

Assumption A.2 (Concept-aligned reconstruction with orthogonal errors). *The reconstruction admits a concept-aligned decomposition $\hat{h}^\Theta = \sum_{j=1}^J \hat{m}_j^\Theta$ (any decoder bias is absorbed into the estimates), and the per-concept reconstruction errors are mutually uncorrelated:*

$$\mathbb{E}[\langle m_j - \hat{m}_j^\Theta, m_k - \hat{m}_k^\Theta \rangle] = 0 \quad \text{for } j \neq k.$$

Assumption A.2 holds, for example, when the concept manifolds occupy mutually orthogonal subspaces and each estimate $\hat{m}_j^\Theta$ and its error lie in the corresponding subspace, or when the errors are independent across concepts. It is an idealization: real concept components overlap, and a trained SAE does not come with a canonical per-concept decomposition of its reconstruction. The assumption serves to isolate the incentive structure of the loss; the experiment of Section A.3 provides the empirical counterpart on real activations.

For each concept, define the *captured energy fraction*

$$C_j(\Theta) = 1 - \frac{\mathbb{E}[\|m_j - \hat{m}_j^\Theta\|_2^2]}{E_j} \leq 1, \quad (5)$$

with $C_j(\Theta) \geq 0$ whenever the estimate is at least as accurate as the trivial estimate $\hat{m}_j^\Theta = 0$.

A.2 ENERGY-WEIGHTED DECOMPOSITION OF THE RECONSTRUCTION LOSS

Proposition A.1 (Energy bias of reconstruction objectives). *Under Assumptions A.1–A.2,*

$$\mathcal{L}_{\text{rec}}(\Theta) = \sum_{j=1}^J E_j (1 - C_j(\Theta)). \quad (6)$$

Equivalently, minimizing the reconstruction loss is equivalent to maximizing the energy-weighted capture

$$G_{\text{rec}}(\Theta) = \sum_{j=1}^J E_j C_j(\Theta). \quad (7)$$

Proof. Write $h - \hat{h}^\Theta = \sum_{j=1}^J (m_j - \hat{m}_j^\Theta)$ using Eq. equation 3 and Assumption A.2. Expanding the squared norm,

$$\mathcal{L}_{\text{rec}}(\Theta) = \sum_{j=1}^J \mathbb{E}[\|m_j - \hat{m}_j^\Theta\|_2^2] + \sum_{j \neq k} \mathbb{E}[\langle m_j - \hat{m}_j^\Theta, m_k - \hat{m}_k^\Theta \rangle].$$

The cross terms vanish by Assumption A.2, and each diagonal term equals $E_j(1 - C_j(\Theta))$ by Eq. equation 5, which gives Eq. equation 6. Since $\mathcal{L}_{\text{rec}}(\Theta) = \sum_j E_j - G_{\text{rec}}(\Theta)$ and $\sum_j E_j$ does not depend on Θ , minimizing the former maximizes the latter. $\square$

Corollary A.1 (Marginal capacity is allocated by energy). *Consider a marginal change of the extractor that improves the capture fractions by $(\Delta C_1, \dots, \Delta C_J)$. The induced reconstruction gain is*

$$\Delta G_{\text{rec}} = \sum_{j=1}^J E_j \Delta C_j.$$

In particular, when a unit of capacity can improve either a background concept $\mathcal{M}_b$ by ΔC_b or a forget concept $\mathcal{M}_f$ by ΔC_f , the reconstruction objective prefers the background concept whenever

$$E_b \Delta C_b > E_f \Delta C_f,$$

and, for comparable capture improvements $\Delta C_b \approx \Delta C_f$, the preference is decided by the energies alone.

Remark A.1. Proposition A.1 does not say that the forget concept is unimportant to the model, only that it contributes little to the reconstruction objective compared with broad background structure. PCA is the extreme case: for linear orthogonal projections, maximizing captured energy is exactly the PCA criterion, so high-variance background directions are selected by construction. NMF and reconstruction-only SAEs are softer versions of the same incentive. This is the mechanism addressed by the contrastive objective of SCALPEL: Proposition 5.1 (proved in Appendix B) shows that adding contrastive learning to reconstruction and sparsity favors sparse representations with stronger within-target alignment and lower target–background overlap.

A.3 MEASURING THE ENERGY GAP ON REAL ACTIVATIONS

We now describe in full the experiment summarized in Section 3 and Figure 2, which measures the energy gap between background structure and target-specific structure on residual-stream activations extracted at layer -4 of Qwen.

Background PCA. Let $H_{\text{bg}} = H_r \cup H_w$ be the union of retain and Wiki activations and H_f the forget activations. We estimate the background mean and covariance using only the background data,

$$\bar{h}_{\text{bg}} = \frac{1}{|H_{\text{bg}}|} \sum_{h \in H_{\text{bg}}} h, \quad \Sigma_{\text{bg}} = \frac{1}{|H_{\text{bg}}|} \sum_{h \in H_{\text{bg}}} (h - \bar{h}_{\text{bg}})(h - \bar{h}_{\text{bg}})^\top.$$

Let $\lambda_1^{\text{bg}} \geq \lambda_2^{\text{bg}} \geq \lambda_3^{\text{bg}} \geq \dots$ be the eigenvalues of Σ_{bg} and $U_{\text{bg},3} \in \mathbb{R}^{d \times 3}$ its first three eigenvectors. The left panel of Figure 2 shows the background activations in these coordinates, $y_{\text{bg}} = U_{\text{bg},3}^\top (h - \bar{h}_{\text{bg}})$ for $h \in H_{\text{bg}}$.

Background-orthogonal forget residuals. We remove the dominant background component from every forget activation,

$$\tilde{h}_f = (I - U_{\text{bg},3} U_{\text{bg},3}^\top)(h_f - \bar{h}_{\text{bg}}), \quad h_f \in H_f.$$

Projecting these residuals back onto $U_{\text{bg},3}$ gives zero coordinates by construction, so visualizing the remaining forget structure requires a second basis: we fit a second PCA directly on $\tilde{H}_f = \{\tilde{h}_f : h_f \in H_f\}$. With $\bar{\tilde{h}}_f$ the mean of the residuals, define

$$\Sigma_f^{\perp \text{bg}} = \frac{1}{|\tilde{H}_f|} \sum_{\tilde{h}_f \in \tilde{H}_f} (\tilde{h}_f - \bar{\tilde{h}}_f)(\tilde{h}_f - \bar{\tilde{h}}_f)^\top,$$

with eigenvalues $\lambda_i^{f \perp \text{bg}}$. The right panel of Figure 2 shows the residual forget activations in the top-3 eigenvectors of $\Sigma_f^{\perp \text{bg}}$. The two panels therefore use two different PCA bases and must be compared through their scale and explained energy, not their absolute coordinates.

Energy comparison. We measure the scale of each displayed manifold by the sum of its first three PCA eigenvalues,

$$\hat{E}_{\text{bg}}^{(3)} = \sum_{i=1}^3 \lambda_i^{\text{bg}}, \quad \hat{E}_f^{\perp \text{bg},(3)} = \sum_{i=1}^3 \lambda_i^{f \perp \text{bg}},$$

and obtain

$$\widehat{E}_{\text{bg}}^{(3)} = 3,787,695, \quad \widehat{E}_f^{\perp \text{bg}, (3)} = 11,517, \quad \frac{\widehat{E}_{\text{bg}}^{(3)}}{\widehat{E}_f^{\perp \text{bg}, (3)}} \approx 329.$$

What the experiment does and does not show. The experiment does not claim that the complete forget manifold carries exactly $329\times$ less energy: it measures the scale of the forget structure that *remains* after removing the three dominant background directions, which is the component a targeted intervention must isolate. The size of the gap nevertheless supports the energy-bias argument of Proposition A.1: the target-specific component is orders of magnitude smaller than the background structure that a reconstruction-based extractor is primarily rewarded for capturing.

B SPARSE CONTRASTIVE LEARNING PROMOTES TARGET-SELECTIVE FEATURES

We give a geometric account of why the combination of reconstruction, sparsity, and contrastive learning used by SCALPEL favors target-selective sparse features. The result characterizes the geometry preferred by the objective when such a representation is compatible with reconstruction; it is not a global convergence result for SAE training.

B.1 SETTING

For a standardized residual activation h_i , the SAE produces non-negative activations $z(h_i) \in \mathbb{R}_+^Q$ and reconstructs

$$\widehat{h}_i = b_D + \sum_{q=1}^Q z_q(h_i) d_q.$$

The contrastive loss operates on the norm-aware code

$$\widetilde{z}_q(h_i) = z_q(h_i) \|d_q\|_2, \quad \widetilde{z}_i = (\widetilde{z}_1(h_i), \dots, \widetilde{z}_Q(h_i)).$$

For every non-zero code, define

$$\rho_i = \|\widetilde{z}_i\|_2, \quad u_i = \frac{\widetilde{z}_i}{\rho_i}.$$

Thus

$$u_i \in \mathbb{R}_+^Q, \quad \|u_i\|_2 = 1,$$

and the cosine similarity used by the contrastive objective is simply

$$\text{sim}(i, j) = \langle u_i, u_j \rangle.$$

We assume that decoder scales are controlled:

$$0 < d_{\min} \leq \|d_q\|_2 \leq d_{\max} < \infty. \quad (\text{A1})$$

This removes the reciprocal rescaling ambiguity between encoder activations and decoder directions.

For anchor i , let $P(i)$ and $N(i)$ be its positive and negative sets, with $p_i = |P(i)|$ and $n_i = |N(i)|$. In SCALPEL, positives belong to the same target concept and negatives are drawn from the background. The multi-positive InfoNCE loss is

$$\ell_i = -\log \frac{\sum_{j \in P(i)} e^{\langle u_i, u_j \rangle / \tau}}{\sum_{j \in P(i)} e^{\langle u_i, u_j \rangle / \tau} + \sum_{j \in N(i)} e^{\langle u_i, u_j \rangle / \tau}}.$$

B.2 RECONSTRUCTION PREVENTS THE TRIVIAL REPRESENTATION

The contrastive direction u_i is undefined when $\widetilde{z}_i = 0$. Reconstruction prevents the all-zero solution from being useful.

Lemma B.1 (Non-triviality from reconstruction). *If $z(h) = 0$ almost surely, the best possible reconstruction is the constant decoder bias*

$$b_D^* = \mathbb{E}[h],$$

with loss

$$L_{\text{const}} = \mathbb{E} [\|h - \mathbb{E}[h]\|_2^2].$$

Consequently, any SAE satisfying

$$L_{\text{rec}} < L_{\text{const}}$$

must use non-zero latent codes on a set of positive probability.

Proof. If $z(h) = 0$, then $\hat{h} = b_D$ for every input. The minimizer of $\mathbb{E}\|h - b_D\|_2^2$ is $b_D = \mathbb{E}[h]$, giving the stated loss. Any reconstruction below this value therefore requires non-zero latent activations. $\square$

Reconstruction therefore supplies a non-trivial representation, while the remaining terms determine how this representation is organized.

B.3 SPARSITY CONCENTRATES THE NORM-AWARE CODE

For a normalized code u , let $u_{(1)} \geq \dots \geq u_{(Q)}$ denote its coordinates in decreasing order.

Lemma B.2 (Sparse energy concentration). *For every $K \geq 1$,*

$$\sum_{q>K} u_{(q)}^2 \leq \frac{\|u\|_1^2}{K},$$

and hence

$$\sum_{q=1}^K u_{(q)}^2 \geq 1 - \frac{\|u\|_1^2}{K}.$$

Moreover,

$$\|u\|_1 \geq \|u\|_2 = 1,$$

with equality if and only if u is one-sparse.

Proof. Since the coordinates are sorted,

$$u_{(K+1)} \leq \frac{\|u\|_1}{K}.$$

Therefore

$$\sum_{q>K} u_{(q)}^2 \leq u_{(K+1)} \sum_{q>K} u_{(q)} \leq \frac{\|u\|_1^2}{K}.$$

The second result follows from $\|u\|_2^2 = 1$. Finally, $\|u\|_1 \geq \|u\|_2$, with equality for a non-negative vector only when at most one coordinate is non-zero. $\square$

Under Assumption (A1), sparsifying z also sparsifies $\tilde{z}$, since multiplication by bounded positive decoder norms does not change its support. Thus the ℓ_1 term favors concentrating the representation on a small number of coordinates, while reconstruction prevents complete collapse.

B.4 CONTRASTIVE LEARNING ALIGNS TARGETS AND SEPARATES BACKGROUND

We now characterize the geometry preferred by the contrastive term.

Proposition B.1 (Optimal contrastive geometry). *For every valid anchor i ,*

$$\ell_i \geq \log \left(1 + \frac{n_i}{p_i} e^{-1/\tau} \right).$$

Equality holds if and only if

$$\langle u_i, u_j \rangle = 1 \quad \forall j \in P(i),$$

and

$$\langle u_i, u_j \rangle = 0 \quad \forall j \in N(i).$$

Therefore, whenever this geometry is compatible with reconstruction, the contrastive objective favors identical normalized codes within a target and disjoint supports between that target and its background negatives.

Proof. Because u_i, u_j are non-negative unit vectors,

$$0 \leq \langle u_i, u_j \rangle \leq 1.$$

Define

$$A_i^+ = \sum_{j \in P(i)} e^{\langle u_i, u_j \rangle / \tau}, \quad A_i^- = \sum_{j \in N(i)} e^{\langle u_i, u_j \rangle / \tau}.$$

Then

$$A_i^+ \leq p_i e^{1/\tau}, \quad A_i^- \geq n_i.$$

Since

$$\ell_i = \log \left(1 + \frac{A_i^-}{A_i^+} \right),$$

we obtain

$$\ell_i \geq \log \left(1 + \frac{n_i}{p_i} e^{-1/\tau} \right).$$

Equality requires simultaneously

$$\langle u_i, u_j \rangle = 1 \quad (j \in P(i))$$

and

$$\langle u_i, u_j \rangle = 0 \quad (j \in N(i)).$$

For unit vectors, the first condition implies $u_i = u_j$. Because all coordinates are non-negative, the second condition implies

$$\text{supp}(u_i) \cap \text{supp}(u_j) = \emptyset.$$

Thus same-target examples share a normalized representation, whereas target and background negatives use disjoint coordinates. $\square$

Combined with Lemma B.2, this gives the preferred geometry of the joint objective: reconstruction keeps the representation non-trivial, sparsity concentrates it onto few coordinates, and the contrastive term encourages those coordinates to be shared by examples of the same target while remaining inactive on background examples.

Importantly, the proposition only separates pairs defined as negatives. If two forget targets are not contrasted against each other, they may share coordinates. This is consistent with the shared features observed empirically in SCALPEL.

B.5 NEAR-OPTIMAL CONTRASTIVE LOSS IMPLIES NEAR-OPTIMAL GEOMETRY

The previous result describes the exact optimum. We can also quantify how approaching this optimum forces the representations toward the same geometry.

Let

$$R_i = \frac{A_i^-}{A_i^+}, \quad R_i^* = \frac{n_i}{p_i} e^{-1/\tau}.$$

Suppose

$$R_i \leq (1 + \delta) R_i^*, \quad \delta \geq 0. \tag{A2}$$

Proposition B.2 (Approximate contrastive separation). *Under Eq. (A2),*

$$\frac{1}{n_i} \sum_{j \in N(i)} \langle u_i, u_j \rangle \leq \tau \delta,$$

and

$$\frac{1}{p_i} \sum_{j \in P(i)} (1 - \langle u_i, u_j \rangle) \leq \frac{\delta}{(1 + \delta)(1 - e^{-1/\tau})}.$$

Consequently,

$$\frac{1}{p_i} \sum_{j \in P(i)} \|u_i - u_j\|_2^2 \leq \frac{2\delta}{(1 + \delta)(1 - e^{-1/\tau})}.$$

Hence, as the contrastive loss approaches its optimum, positive codes collapse toward one another while their average similarity with background negatives approaches zero.

Proof. Observe that

$$\frac{R_i}{R_i^*} = \underbrace{\frac{A_i^-}{n_i}}_{\geq 1} \underbrace{\frac{p_i e^{1/\tau}}{A_i^+}}_{\geq 1}.$$

If their product is at most $1 + \delta$, each factor is itself at most $1 + \delta$.

For the negatives,

$$\frac{1}{n_i} \sum_{j \in N(i)} e^{\langle u_i, u_j \rangle / \tau} \leq 1 + \delta.$$

Using $e^x \geq 1 + x$ gives

$$\frac{1}{n_i} \sum_{j \in N(i)} \langle u_i, u_j \rangle \leq \tau \delta.$$

For the positives,

$$\frac{1}{p_i} \sum_{j \in P(i)} e^{-(1 - \langle u_i, u_j \rangle) / \tau} \geq \frac{1}{1 + \delta}.$$

For $x \in [0, 1]$, concavity of $1 - e^{-x/\tau}$ gives

$$1 - e^{-x/\tau} \geq (1 - e^{-1/\tau})x.$$

Therefore,

$$(1 - e^{-1/\tau}) \frac{1}{p_i} \sum_{j \in P(i)} (1 - \langle u_i, u_j \rangle) \leq 1 - \frac{1}{1 + \delta} = \frac{\delta}{1 + \delta},$$

which proves the second inequality. Finally,

$$\|u_i - u_j\|_2^2 = 2(1 - \langle u_i, u_j \rangle),$$

giving the last result. $\square$

Since

$$\ell_i = \log(1 + R_i),$$

condition (A2) is equivalently a bound on the excess contrastive loss: $\delta \rightarrow 0$ whenever $\ell_i \rightarrow \ell_i^*$.

B.6 CONNECTION TO THE SELECTIVITY SCORE

Recall the SCALPEL score

$$s_q(G_a) = \frac{\mu_q(G_a)}{\mu_q(\text{bg}) + \epsilon},$$

where

$$\mu_q(G_a) = \mathbb{E}_{i \in G_a} [\tilde{z}_q(h_i)], \quad \mu_q(\text{bg}) = \mathbb{E}_{j \in \text{bg}} [\tilde{z}_q(h_j)].$$

At the ideal sparse contrastive geometry, suppose target G_a uses coordinate q . Reconstruction ensures a non-zero code scale, sparsity concentrates the target representation on q , and Proposition B.1 gives

$$\tilde{z}_q(h_i) > 0 \quad \text{for } i \in G_a, \quad \tilde{z}_q(h_j) = 0 \quad \text{for } j \in \text{bg}.$$

Hence

$$\mu_q(G_a) > 0, \quad \mu_q(\text{bg}) = 0,$$

and therefore

$$s_q(G_a) = \frac{\mu_q(G_a)}{\epsilon}.$$

Thus the geometry favored by the joint objective is exactly the geometry rewarded by the selectivity criterion: the sparsity term concentrates target information onto few coordinates, while the contrastive term increases agreement within the target and suppresses overlap of those coordinates with background examples. Proposition B.2 further shows that this behavior emerges continuously as the contrastive loss approaches its optimum.

C SELECTIVITY AND INTERVENTION SAFETY

This appendix proves Proposition 5.2 of the main text: the selectivity score used by SCALPEL controls the expected magnitude of the perturbation applied to background activations. Together with the sparse contrastive analysis of Appendix B, it completes the training-to-intervention argument: contrastive learning favors sparse representations with stronger within-target alignment and lower target–background overlap, while selecting high-score features limits their expected footprint on retain and Wiki representations under intervention.

C.1 SETTING

We use the notation of Sections 4.3 and 4.5. For a standardized activation $\bar{h}$, the encoder produces a non-negative sparse code $z(\bar{h}) \in \mathbb{R}_+^Q$, so that

$$z_q(\bar{h}) \geq 0 \quad \text{for all } q, \bar{h}, \quad (8)$$

and $d_q \in \mathbb{R}^d$ is the decoder direction of feature q . For an author-level target G_a , with $\bar{\mathcal{H}}_{\text{bg}} = \bar{\mathcal{H}}_r \cup \bar{\mathcal{H}}_w$ the standardized background activations, recall the mean norm-aware activations and the selectivity score (Eq. equation 1 of the main text),

$$\mu_q(G_a) = \mathbb{E}_{\bar{h} \sim \bar{\mathcal{H}}_{G_a}} [z_q(\bar{h}) \|d_q\|_2], \quad \mu_q(\text{bg}) = \mathbb{E}_{\bar{h} \sim \bar{\mathcal{H}}_{\text{bg}}} [z_q(\bar{h}) \|d_q\|_2], \quad s_q(G_a) = \frac{\mu_q(G_a)}{\mu_q(\text{bg}) + \epsilon},$$

with $\epsilon > 0$. SCALPEL selects $S_{G_a} = \{q : s_q(G_a) \geq s_{\min}\}$ for a threshold $s_{\min} > 0$, and the intervention applies the perturbation

$$\delta_{G_a}(\bar{h}) = \alpha \sum_{q \in S_{G_a}} z_q(\bar{h}) d_q, \quad \alpha \geq 0. \quad (9)$$

C.2 THE PERTURBATION BOUND

Proposition C.1 (Selectivity controls background perturbation; Proposition 5.2 of the main text). *If every selected feature satisfies $s_q(G_a) \geq s_{\min}$, then*

$$\mathbb{E}_{\bar{h} \sim \bar{\mathcal{H}}_{\text{bg}}} [\|\delta_{G_a}(\bar{h})\|_2] \leq \alpha \sum_{q \in S_{G_a}} \mu_q(\text{bg}) \leq \frac{\alpha}{s_{\min}} \sum_{q \in S_{G_a}} \mu_q(G_a) - \alpha \epsilon |S_{G_a}| \leq \frac{\alpha}{s_{\min}} \sum_{q \in S_{G_a}} \mu_q(G_a). \quad (10)$$

Proof. By the triangle inequality and the non-negativity of the codes (Eq. equation 8), for every $\bar{h}$,

$$\|\delta_{G_a}(\bar{h})\|_2 \leq \alpha \sum_{q \in S_{G_a}} z_q(\bar{h}) \|d_q\|_2.$$

Taking the expectation over $\bar{h} \sim \bar{\mathcal{H}}_{\text{bg}}$ and using linearity gives the first inequality of Eq. equation 10. For the second, each selected feature satisfies $s_q(G_a) = \mu_q(G_a)/(\mu_q(\text{bg}) + \varepsilon) \geq s_{\min}$, hence

$$\mu_q(\text{bg}) \leq \frac{\mu_q(G_a)}{s_{\min}} - \varepsilon.$$

Summing over $q \in S_{G_a}$ and multiplying by α yields the second inequality; dropping the non-negative term $\alpha\varepsilon|S_{G_a}|$ yields the third. $\square$

C.3 DISCUSSION

Remark C.1 (Interpretation and monotonicity). The quantity $\sum_{q \in S_{G_a}} \mu_q(G_a)$ is the total norm-aware activation mass of the selected features on the target: it is the natural scale of the intervention, since the same triangle-inequality argument applied to target activations gives $\mathbb{E}_{\bar{h} \sim \bar{\mathcal{H}}_{G_a}} [\|\delta_{G_a}(\bar{h})\|_2] \leq \alpha \sum_{q \in S_{G_a}} \mu_q(G_a)$. Proposition C.1 states that the corresponding bound on *background* data is smaller by the factor $1/s_{\min}$: relative to the activation mass they carry on the target, high-selectivity features can only perturb background activations weakly. The bound is monotone in the threshold, raising $s_{\min}$ tightens it, and the intermediate form in Eq. equation 10 shows that the constant ε in the score buys an additional slack of $\alpha\varepsilon$ per selected feature.

Remark C.2 (Standardized versus original residual space). The bound is stated in standardized activation space, where the SAE operates (Section 4.2), while the intervention of Eq. equation 2 is applied to the residual stream. If the decoded contribution is mapped back to the original space through the fixed standardization, the perturbation becomes $\alpha \sum_{q \in S_{G_a}} z_q(\bar{h}) (\sigma_H \odot d_q)$, and Proposition C.1 holds verbatim with $\|d_q\|_2$ replaced by $\|\sigma_H \odot d_q\|_2 \leq \|\sigma_H\|_\infty \|d_q\|_2$ in the definition of μ_q ; in particular the same bound holds up to the fixed factor $\|\sigma_H\|_\infty$. The selectivity ratio $s_q(G_a)$, and hence the $1/s_{\min}$ gain, is unchanged by any per-feature rescaling of $\|d_q\|_2$, since the rescaling appears in both the numerator and the denominator of the score.

Remark C.3 (What the bound does and does not guarantee). Proposition C.1 is a *representation-level safety bound*, and its scope should be read precisely. First, it controls the *expected* perturbation magnitude over background activations, not the worst case: individual background tokens on which a selected feature fires strongly can still be perturbed substantially. Second, it aggregates over the selected set, so the bound grows with $|S_{G_a}|$; selecting few, highly selective features is favored on both counts. Third, and most importantly, it says nothing about model *behavior*: relating a perturbation at layer ℓ to a change in the output distribution would require additional assumptions on the downstream layers. Fourth, the score measures activation magnitude and decoder norm but not the *orientation* of d_q relative to the local geometry of the forget and background manifolds; two features with similar scores can therefore produce different behavioral effects, a variability that Section 7.3 measures directly and that a geometry-aware score could reduce at substantially higher cost (Appendix I).

Remark C.4 (Connection to the training analysis). Appendix B shows that the contrastive objective favors sparse representations with stronger within-target alignment and lower target-background overlap, thereby promoting features with higher selectivity scores $s_q(G_a)$. Proposition C.1 completes the argument: selecting features above a threshold $s_{\min}$ limits their expected perturbation on background representations relative to their activation on the target.

D UMAP : MAINFOLD OF THE DATASETS

E HYPERPARAMETER OPTIMIZATION PROTOCOL AND SEARCH SPACES

This appendix specifies the complete model-selection protocol. It has two levels. **Level 1** is the hyperparameter search: one independent Optuna study (Akiba et al., 2019) per (model $\times$ method) pair, 3 models $\times$ (5 SAE-family methods + 2 trainer methods) = 21 studies, all run with a 5 seeds

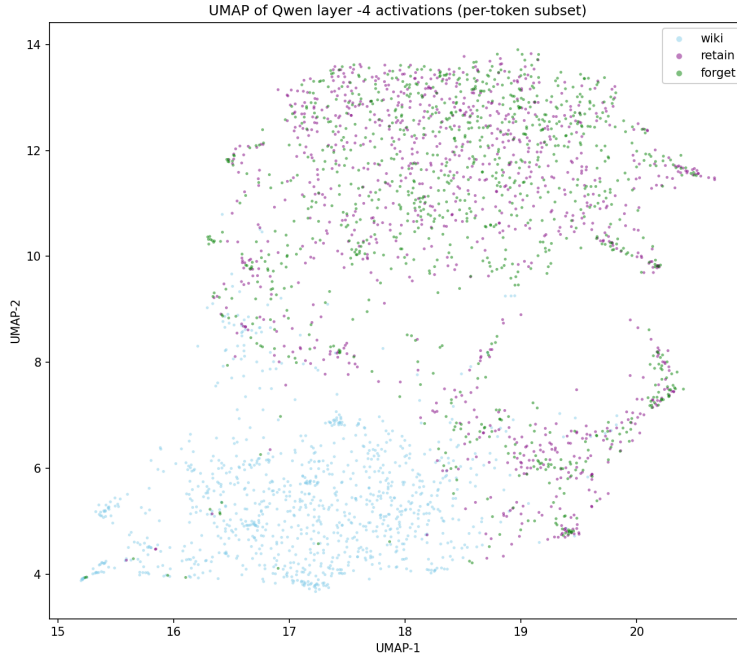


Figure 5: UMAP projection (McInnes et al., 2018) of residual-stream activations from the forget, retain, and Wiki sets. Forget and retain examples strongly overlap because they come from the same generation process and differ only through the random assignment of authors to the splits: the full forget set is not a globally separable concept in activation space.

(0-4). A study never samples over methods: each method is pinned to its own study and therefore receives its own full trial budget. **Level 2** freezes the winning configuration of each study and reruns the complete pipeline over five independent seeds; its essentials are summarized in Section E.5 below, and the reported results come exclusively from Level 2.

E.1 SELECTION OBJECTIVE

All 21 studies maximize the same scalar, the *retain-forget margin*

$$\text{margin} = \frac{1}{2}(P_{\text{retain}} + P_{\text{holdout}}) - P_{\text{forget}}, \quad (11)$$

where P_{forget} , P_{retain} , and P_{holdout} are teacher-forced ground-truth answer probabilities on OpenUnlearning’s own probe sets (*forget05* and *holdout05*; probe batch size 32; parity with the OpenUnlearning data verified by a dedicated check). Model selection never uses generation-based or test metrics, so the selection signal is decoupled from the reported evaluation. An earlier constrained objective (smallest α such that P_{forget} falls below a threshold) is kept in the codebase for diagnostics only; all reported studies use the margin of Eq. equation 11.

Operating point within a trial (SAE track). For representation-based methods, the free knob inside a trial is the intervention strength α . Each trial trains the extractor, selects features, then evaluates a coarse grid $\alpha \in \{0, 2, 4, 8, 16, 32\}$ in a single prefix-shared forward pass, followed by golden-section refinement (ratio 0.618) on the bracket around the coarse argmax, with a budget of at most 8 refinement evaluations and a relative resolution of 0.05. The trial value is the best margin over all evaluated α . This fast, margin-maximizing search is used only for model selection; it is distinct from the log-space bisection of Appendix H, which traces the full trade-off curves reported in the benchmark.

Operating point within a trial (trainer track). GradDiff and RMU have no α ; the free knob is the training checkpoint. Every checkpoint (stride 1) is scored with the same margin, and the trial value is the maximum over *non-degenerate* checkpoints. A degeneracy guard rejects checkpoints

whose forget probability falls below $0.1 \times$ the retain-reference threshold as collapsed rather than unlearned (the threshold is computed from a retain-only finetuned reference); if all checkpoints are degenerate, the trial receives a penalty of -10^9 .

E.2 SEARCH SPACES

Table 1 gives the SAE-family search space and Table 2 the trainer search space; the distributions below are exactly those recorded in the study databases.

Table 1: SAE-family search space (per trial). “Trained SAEs” means all methods except NMF; “contrastive family” means all contrastive variants.

Hyperparameter	Range / choices	Sampling	Applies to
Latent dimension Q	{64, 512, 1024, 2048, 4096, 8192}	categorical	all
Selection threshold $s_{\min}$	[1.0, 100.0]	uniform	all
Batch size	{64, 128, 256, 512}	categorical	trained SAEs
Training epochs	[5, 40]	int. uniform	trained SAEs
Encoder depth	{1, 2}	categorical	trained SAEs
Sparsity coefficient λ	$[10^{-4}, 10^{-1}]$	log-uniform	trained SAEs
Temperature τ	{0.1, 0.5}	categorical	contrastive family
Contrastive weight β	$[10^{-4}, 1.0]$	log-uniform	contrastive SAE
Common-block fraction	{0.25, 0.5, 0.75}	categorical	Split
$\beta_{\text{com}}, \beta_{\text{diff}}$	$[10^{-4}, 1.0]$ each	log-uniform	Split
NMF max iterations	[50, 500]	int. uniform	NMF

Fixed (not searched), SAE family: learning rate 10^{-3} ; extraction layer -4 (fourth from last); Wiki reconstruction weight $\lambda_{\text{wiki}} = 1.0$, so the training loss is $\text{MSE}(\text{TOFU}) + \lambda_{\text{wiki}} \text{MSE}(\text{Wiki}) + \lambda \mathcal{L}_{\text{sparse}} + \text{contrastive terms}$; author-level contrastive grouping (Section 4.1); per-dimension z-score input normalization computed once from the TOFU token cache and frozen across trials; feature selection by the per-author norm-aware forget/background ratio of Eq. equation 1 with top-1 fallback, using at most 20,000 tokens per side.

Table 2: Trainer search space (GradDiff and RMU). The names `gamma` and `alpha` are the trainer implementation’s loss weights and are unrelated to the paper’s intervention strength α and norm exponent γ .

Hyperparameter	Range / choices	Sampling	Method
Learning rate	$[10^{-6}, 10^{-4}]$	log-uniform	both
Training epochs	{10, 15, 20, 25, 30}	categorical	both
Forget-loss weight (<code>gamma</code>)	[0.1, 5.0]	log-uniform	both
Retain-loss weight (<code>alpha</code>)	[0.1, 5.0]	log-uniform	both
Retain loss type	{NLL, KL}	categorical	GradDiff
Steering coefficient	[0.5, 40.0]	log-uniform	RMU

Fixed, trainers: per-device train batch size 4; RMU retain loss `EMBED_DIFF`; checkpoints saved per epoch (model weights only); OpenUnlearning’s generation-based evaluation disabled during training. The epoch grid was raised from the default {2, 4, 6, 8, 10} to {10, ..., 30} because the retain utility of GradDiff and RMU is non-monotonic in training and recovers late.

E.3 SEARCH ALGORITHM, BUDGET, AND PRUNING

All studies use Optuna’s TPE sampler with seed 0, direction *maximize*. SAE-family studies additionally use a median pruner (3 startup trials): the pruning signal reported before refinement is the best margin on the coarse α grid, so a pruned trial skips the golden-section refinement but its record is still written. Trainer studies report the same intermediate value but never prune, since training has already happened by scoring time. The budget is 25 trials per SAE-family study and 20 per trainer study; the budget is a total target, and resubmitted jobs resume the persistent study database until it is reached, which makes walltime-limited cluster jobs safe. For cost control, the

base model and the ~ 25 GB activation cache are loaded once per study rather than once per trial. A built-in uniform-random searcher over the same spaces exists as a fallback, but all reported results use Optuna/TPE.

E.4 STUDY OUTCOMES AND WINNING CONFIGURATIONS

Table 3 reports, for every study, the number of completed and pruned trials (pruned trials consume budget) and the best margin found, as of the reported snapshot; resumable studies below their trial target are marked by their counts.

Table 3: Per-study outcomes: completed trials / pruned trials / best margin (Eq. equation 11).

Method	Qwen2.5	Llama3.2	Gemma2
Contrastive SAE	7 / 18 / 0.528	16 / 9 / 0.532	7 / 18 / 0.518
SCALPEL-Split	17 / 8 / 0.526	11 / 14 / 0.533	9 / 15 / 0.517
Standard SAE	15 / 10 / 0.073	15 / 8 / 0.088	12 / 13 / 0.313
NMF	20 / 5 / 0.020	18 / 7 / ≈ 0	11 / 14 / 0.020
GradDiff	20 / 0 / 0.509	20 / 0 / 0.517	20 / 0 / 0.527
RMU	20 / 0 / 0.531	20 / 0 / 0.526	20 / 0 / 0.524

The selection margins already anticipate the benchmark: reconstruction-only extraction (standard SAE, NMF) cannot reach a useful margin on any model, while the contrastive variants match the strong trainer baselines at selection time.

Each study’s winning configuration is copied verbatim into the experiment configuration by a single generator script; no configuration is hand-edited. As an example, the winning Qwen contrastive-SAE configuration is: latent dimension 1024, encoder depth 2, 19 epochs, batch size 256, sparsity coefficient 2.34×10^{-3} , $\tau = 0.1$, contrastive weight 2.95×10^{-2} , selection threshold 12.7.

One documented post-hoc override. For GradDiff only, the winning epoch count is replaced by a single epoch with sub-epoch checkpointing (a checkpoint every 20 steps, ~ 40 checkpoints, degeneracy floor 0.1): at the selected learning rate GradDiff collapses the forget set inside the first epoch, so per-epoch checkpoints only ever contain already-collapsed models, and sub-epoch checkpointing is required to resolve the transition. This is the only manual deviation from the search winners, and it is applied identically across models and seeds.

E.5 FROM FROZEN CONFIGURATIONS TO THE MULTI-SEED BENCHMARK

With hyperparameters frozen, Level 2 runs five independent end-to-end pipelines per model (fine-tuning seeds 0–4; unlearning and SAE seeds fixed to 0), so the reported error bars cover the full pipeline, not just the unlearning stage. For each seed: (i) the full model and the retain-only reference are finetuned from the base checkpoint (learning rate 5×10^{-5} , 5 base epochs with a balanced $\{-1, 0, +1\}$ epoch jitter dealt round-robin and shuffled deterministically, so no jitter level is missed over five seeds; full and retain share the seed’s epoch count to remain comparable); (ii) the forget side uses all 13,000 augmented forget rows, and the retain side a seed-deterministic random sample of 64,000 rows from the augmented retain pool, while probe and evaluation data are identical across seeds; (iii) a divergence gate verifies that the full and retain models actually separate on the forget set before the seed’s branch runs; (iv) the activation cache (layer -4) and the retain-reference threshold are re-extracted from that seed’s own models; (v) every method retrains with its frozen configuration and re-selects its operating point against that seed’s threshold, SAE-family methods rerun the α search, trainers re-select the checkpoint; and (vi) each winner receives a full OpenUnlearning TOFU evaluation and one row in the results registry. The whole pipeline is emitted as a dependency-ordered SLURM DAG, resumable through per-stage sentinels. Aggregation follows Appendix H: per-method mean $\pm$ population standard deviation across seeds; where a cell has fewer than five completed seeds at the reported snapshot, we state the per-cell n alongside the error bars.

E.6 COMPUTE FOOTPRINT

Each study runs on a single H100 with 64–128 GB of RAM and 6 CPUs; SAE-family studies take 4–24h of walltime (resumed until the trial target), trainer studies 12–24h. Benchmark stages run on smaller MIG slices where they fit (SAE training and α search on a 2g.20gb slice, raised to 3g.40gb for Gemma-2 due to its hidden-size of 2304 and 256k vocabulary). Gemma-2 uses eager attention throughout, as the SDPA soft-capping implementation produced numerical instabilities.

F BENCHMARKS FOR QWEN, LLAMA AND GEMMA

F.1 MODEL UTILITY VS FORGET PROB

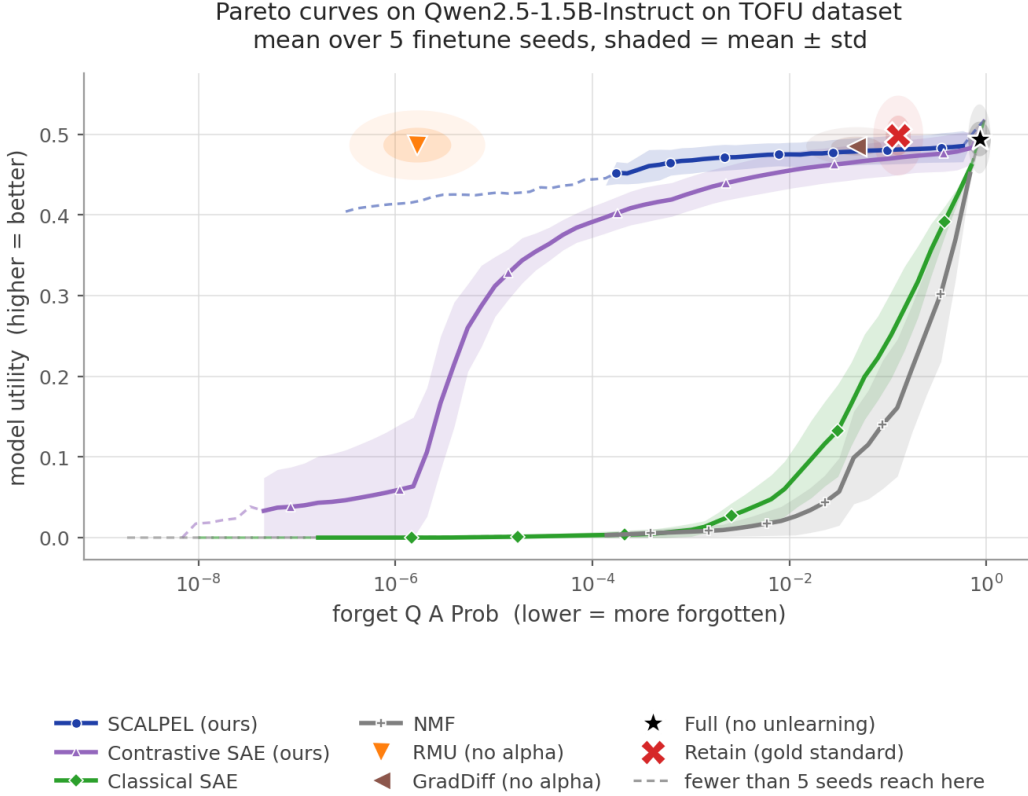


Figure 6: Unlearning trade-off on TOFU forget10 for Qwen2.5-1.5B-Instruct, after method-specific hyperparameter optimization and aggregation over five independent end-to-end seeds. Each curve varies the intervention strength: lower forget probability indicates stronger forgetting, while higher model utility indicates better preservation. Results for Llama and Gemma are reported in Appendix F.1.

F.2 TRADE-OFFS ACROSS MULTIPLE UNLEARNING METRICS

G FEATURE SELECTIVITY ACROSS EXTRACTORS

Figure 12 complements the representation-level geometry of Figure 4a. The contrastive SAE produces more selective top-ranked features than the reconstruction-only SAE, while SCALPEL-Split yields the strongest selectivity. This supports the interpretation that contrastive learning does not merely reorganize the latent geometry globally, but translates this separation into individual features that are more specific to the forget targets.

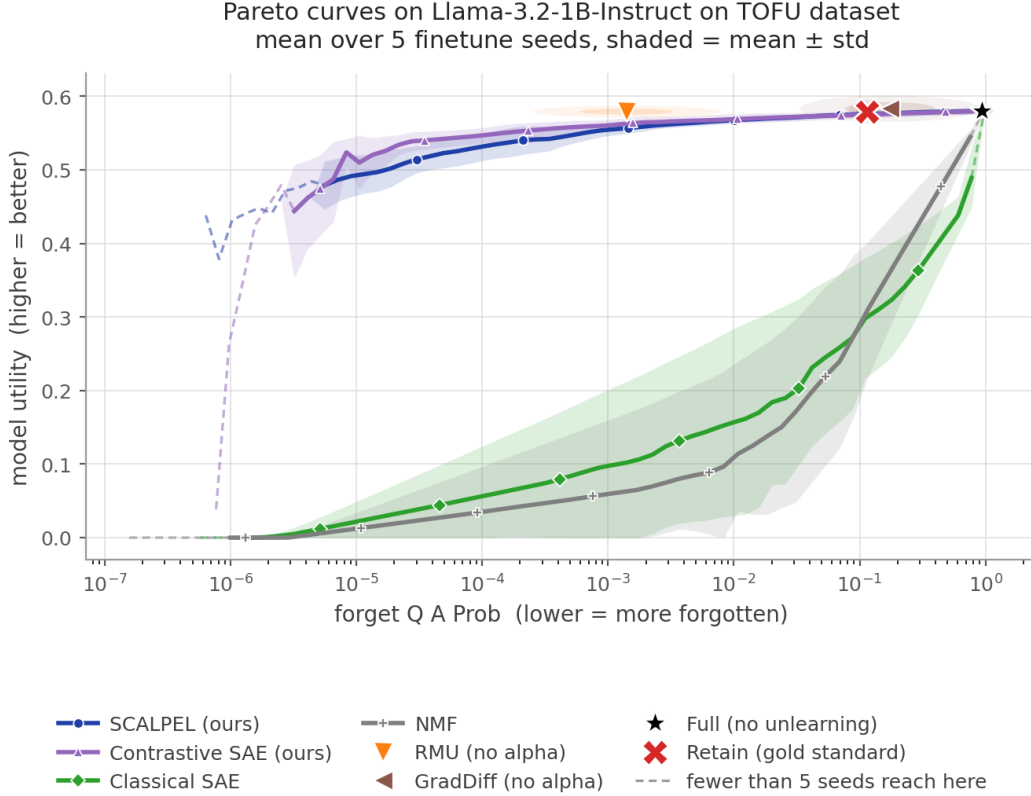


Figure 7: Unlearning trade-off on TOFU forget10 for Llama-3.2-1B-Instruct, after method-specific hyperparameter optimization and aggregation over five independent end-to-end seeds. Each curve varies the intervention strength: lower forget probability indicates stronger forgetting, while higher model utility indicates better preservation.

H ADAPTIVE INTERVENTION-STRENGTH SEARCH

Most methods expose a scalar parameter that controls the strength of unlearning; we denote it generically by α , although its precise meaning depends on the method (intervention strength for representation-based methods, update scale for parameter-based ones). A small value leaves the model close to θ_{full} , while a very large value produces near-complete forgetting together with substantial degradation. A uniform grid over α is inefficient: the practically relevant transition often occupies a narrow interval, and α can span several orders of magnitude. We therefore trace each trade-off curve with an adaptive bisection in $\log \alpha$.

Procedure.

1. **Endpoints.** Evaluate two endpoints α_{low} and α_{high} , chosen to bracket the transition from the nearly unchanged full model to a strongly degraded model. Each evaluation runs the OpenUnlearning pipeline and returns the pair $(P_f(\alpha), P_r(\alpha))$ of forget and retain ground-truth probabilities. All evaluated points are cached, so an interrupted search resumes without recomputation.
2. **Interval selection.** At each iteration, sort the evaluated points by α and consider every adjacent pair (α_i, α_{i+1}) . Prioritize intervals whose forget-probability range overlaps the focus region $\mathcal{B}_f = [0.05, 0.70]$, where the practically relevant part of the trade-off lies; among intervals with the same overlap, select the one with the largest gap in forget probability.

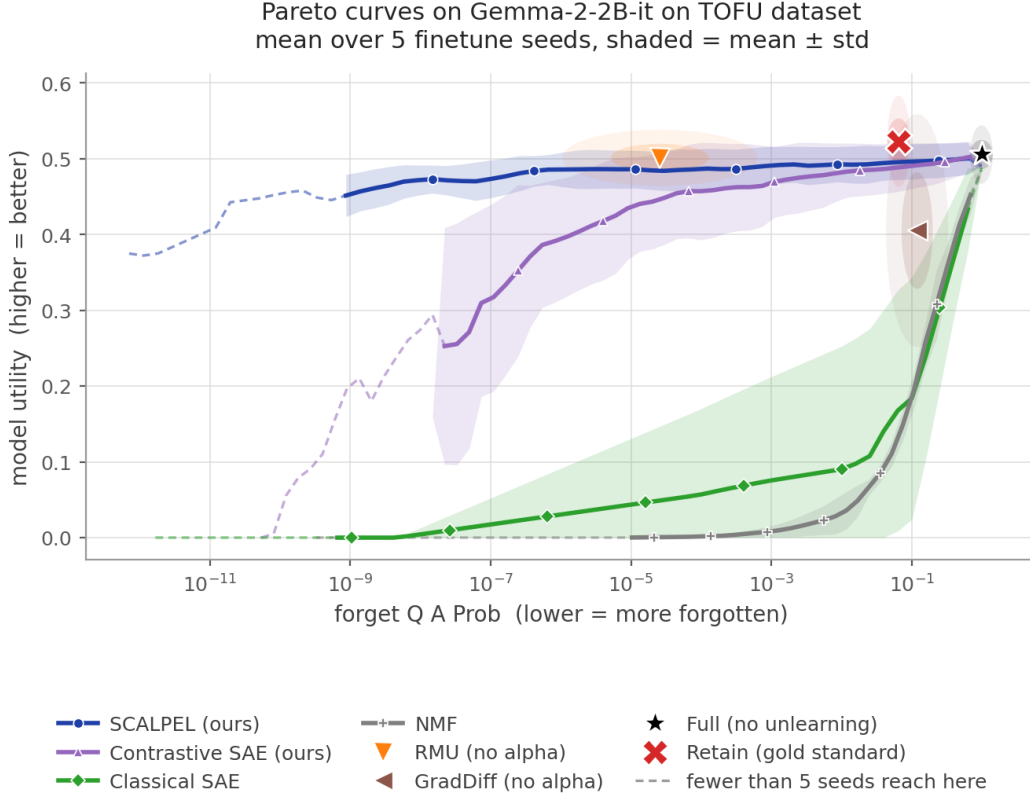


Figure 8: Unlearning trade-off on TOFU forget10 for Gemma-2-2B-it, after method-specific hyperparameter optimization and aggregation over five independent end-to-end seeds. Each curve varies the intervention strength: lower forget probability indicates stronger forgetting, while higher model utility indicates better preservation.

3. **Refinement.** Evaluate the geometric midpoint $\alpha_{\text{new}} = \sqrt{\alpha_i \alpha_{i+1}}$, i.e. the midpoint of the interval in $\log \alpha$ space.
4. **Stopping.** Repeat until the prescribed budget of evaluation points is reached or no interval contains a meaningful unresolved gap.

The procedure keeps the two endpoint evaluations and concentrates the remaining budget in the steep region of the curve, producing a dense estimate of the Pareto front without a large uniform sweep.

Aggregation across seeds. After evaluation, the points of each seed are sorted by α . To aggregate the five seeds, each seed-specific curve is interpolated onto a common forget-probability grid over the seeds’ shared support; the corresponding retain values are then averaged across seeds, and error bars are computed from the five independent runs. This yields comparable mean curves and uncertainty estimates even though the adaptive search evaluates different α values in different seeds.

I EXTENDED LIMITATIONS AND FUTURE WORK

This appendix expands the limitations summarized in Section 8.

Scope of the evaluation. Our experiments cover TOFU and three relatively small models. Although Qwen, Llama, and Gemma represent distinct families, further work is needed to establish whether the same behavior holds for larger models and less controlled datasets. We have not yet completed the evaluation on MUSE, which would require concept groups at the document or chunk

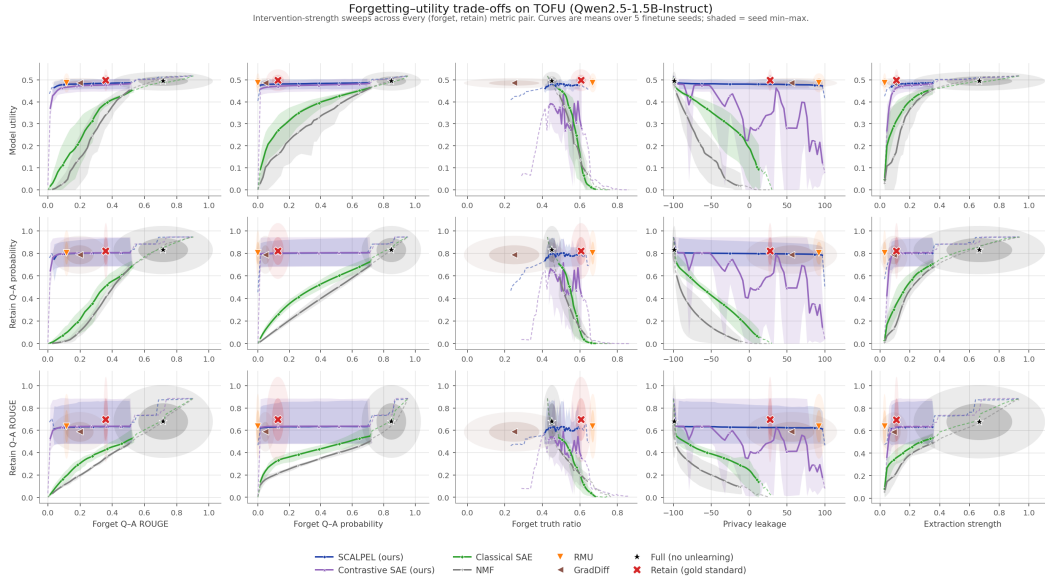


Figure 9: Multi-metric forgetting-utility trade-offs on TOFU forget10 for Qwen2.5-1.5B-Instruct, after method-specific hyperparameter optimization. Curves show the mean over five finetuning seeds, and shaded regions show the seed-wise minimum-maximum range. Columns correspond to different forget-side metrics, while rows report preservation through model utility, retain Q-A probability, and retain Q-A ROUGE, for which higher values are better. Lower forget Q-A ROUGE and forget Q-A probability indicate stronger forgetting. Forget truth ratio and extraction strength are interpreted relative to the retain-only reference, while privacy leakage is best when close to zero. The black star and red cross denote the full and retain-only reference models, respectively; RMU and GradDiff are shown at their selected operating points.

level rather than at the author level, and on WMDP-Bio. Our theory also predicts where the advantage should shrink: the energy-bias argument (Appendix A) implies that SCALPEL helps most when target concepts carry low energy in representation space, so the gap with existing methods may narrow on datasets where targets already contribute strongly to the activations.

Concept granularity. The author-level definition of a concept fits the structure of TOFU and the right-to-be-forgotten setting, where each person is naturally a separate target. Other applications may need a different granularity, a document, a fact, or a single sample. The anchor-based formulation supports these settings in principle through suitable data augmentation (each anchor’s views providing the positives), but their stability and computational cost remain to be studied. Relatedly, we have not yet measured how the method behaves as the fraction of data to unlearn grows; because the intervention acts locally on selected directions, we expect robustness to this parameter, but it must be verified.

Computational cost and design choices. The method depends on several design choices: the intervention layer, the SAE architecture and latent dimension, the contrastive weight and temperature, the augmentation procedure, the selection threshold, and the strength α . We reduce this dependence through per-method hyperparameter optimization and multi-seed evaluation, but the full protocol is expensive: it requires finetuning full and retain-only references, caching token-level activations, training several sparse dictionaries, and evaluating many intervention strengths. The quality of the augmented data is itself a limitation, we rely on a fixed set of paraphrasing and transformation procedures, and richer augmentation could improve the stability of the extracted features and their coverage of the target concept.

Scope of the theory. The theoretical results are deliberately local and rely on stated idealizations. The energy-bias result assumes centered, uncorrelated concept components and orthogonal per-concept reconstruction errors (Appendix A); the contrastive-selectivity analysis assumes non-

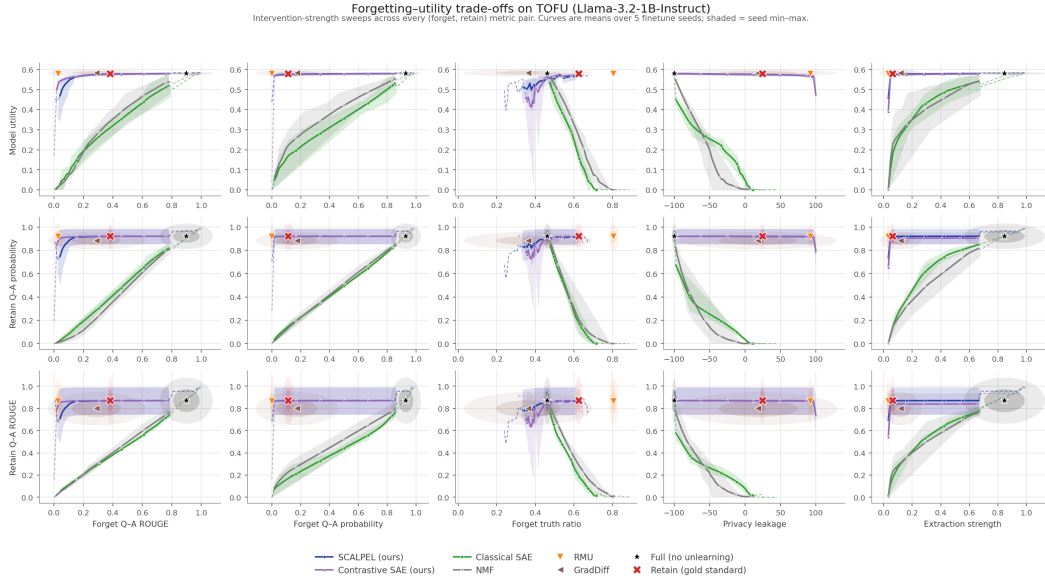


Figure 10: Multi-metric forgetting-utility trade-offs on TOFU forget10 for Llama-3.2-1B-Instruct, after method-specific hyperparameter optimization. Curves show the mean over five finetuning seeds, and shaded regions show the seed-wise minimum-maximum range. Columns correspond to different forget-side metrics, while rows report preservation through model utility, retain Q-A probability, and retain Q-A ROUGE, for which higher values are better. Lower forget Q-A ROUGE and forget Q-A probability indicate stronger forgetting. Forget truth ratio and extraction strength are interpreted relative to the retain-only reference, while privacy leakage is best when close to zero. The black star and red cross denote the full and retain-only reference models, respectively; RMU and GradDiff are shown at their selected operating points.

negative codes, controlled decoder scales, and that sparse target-background separation is compatible with comparable reconstruction and sparsity cost (Appendix B); and the perturbation bound controls only the expected representation-level footprint of the intervention, not downstream behavior (Appendix C). These assumptions make the mechanisms analyzable but do not fully capture the geometry of language-model representations, and none of the results guarantees complete forgetting or behavioral preservation.

Score geometry. The score $s_q(G_a)$ uses activation magnitudes and decoder norms but ignores the orientation of d_q relative to the local forget and background manifolds, which likely explains why features with similar scores can produce different unlearning effects (Section 7.3). A geometry-aware score could compare each decoder direction with local principal directions of the two manifolds, and a geometry-aware intervention could move the activation along, or back toward, the retain manifold instead of applying a straight subtraction, potentially approaching the behavior of a model trained only on retained data. Both require repeated local manifold estimation (e.g. local PCA), making selection and intervention substantially more expensive; we view them as promising extensions rather than free improvements.

Shared features across targets. Several forget authors select the same latent features. This sharing is useful, it means forgetting all ten authors needs far fewer than fifty directions, reducing memory and inference cost, but it may also couple targets: removing a shared feature for one author could affect another author represented by the same coordinate. Further analysis is needed to determine whether these features capture genuinely shared forget structure or introduce unwanted interference.

Inference overhead and sequential unlearning. SCALPEL adds a small inference-time overhead: the SAE encoder and the selected decoder directions must remain available so the intervention can be applied at the chosen layer. Sequential unlearning is an open problem: after unlearning one group, removing a second currently requires either retraining SCALPEL on the combined forget set

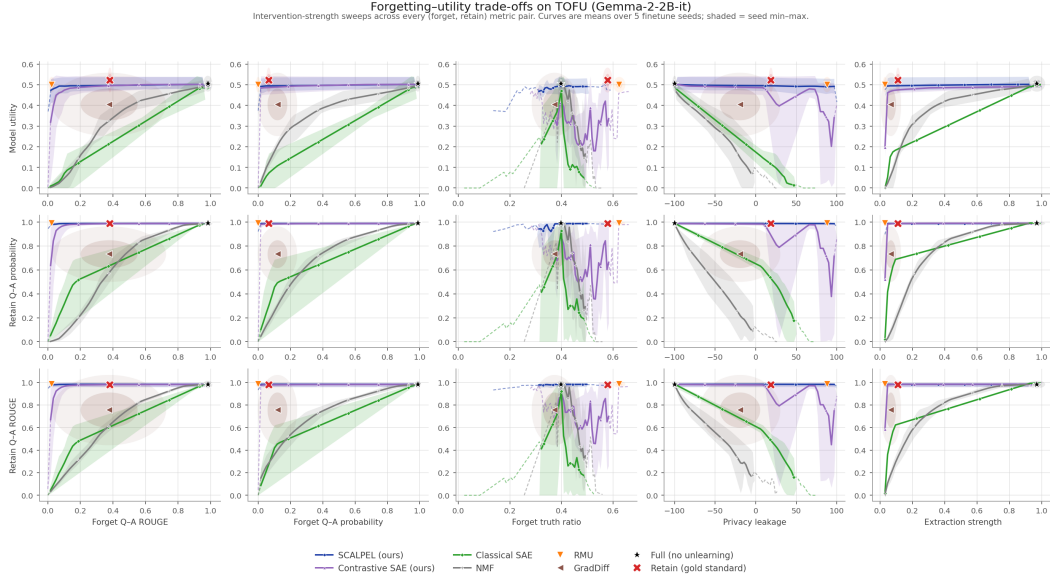


Figure 11: Multi-metric forgetting-utility trade-offs on TOFU forget10 for Gemma-2-2B-it, after method-specific hyperparameter optimization. Curves show the mean over five finetuning seeds, and shaded regions show the seed-wise minimum-maximum range. Columns correspond to different forget-side metrics, while rows report preservation through model utility, retain Q-A probability, and retain Q-A ROUGE, for which higher values are better. Lower forget Q-A ROUGE and forget Q-A probability indicate stronger forgetting. Forget truth ratio and extraction strength are interpreted relative to the retain-only reference, while privacy leakage is best when close to zero. The black star and red cross denote the full and retain-only reference models, respectively; RMU and GradDiff are shown at their selected operating points.

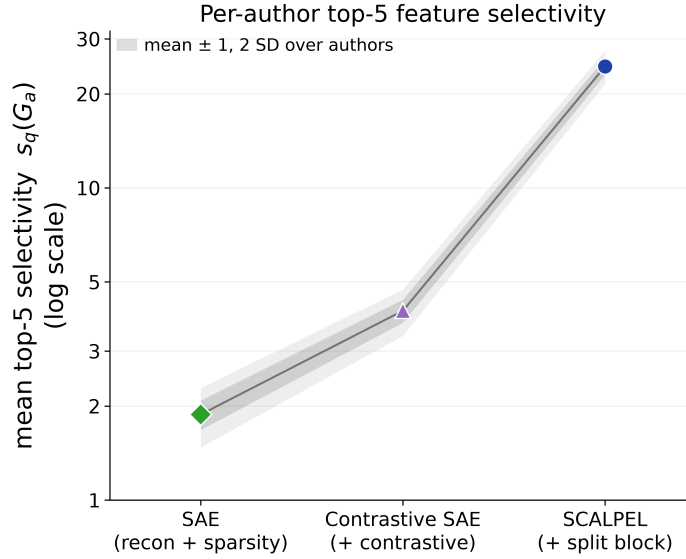


Figure 12: Per-author top-5 feature selectivity for the reconstruction-only SAE, contrastive SAE, and SCALPEL-Split, evaluated across the five Qwen benchmark seeds. Points correspond to the mean top-5 selectivity score $s_q(G_a)$ for each author, while the shaded region summarizes variation across authors. Adding contrastive learning increases feature selectivity over the standard SAE, and the split architecture further amplifies this effect.

or stacking a second intervention, which increases inference cost with every new request. Making the method scale to repeated unlearning requests is an important direction.

Context-aware extraction. The encoder processes token representations individually. These already carry contextual information from earlier transformer layers, but the extraction mechanism does not explicitly model the full sample context; a context-aware encoder could better isolate associations that only become identifiable at the sentence or document level.

Behavioral versus parametric removal. Finally, two limitations concern what “unlearning” means here. First, existing metrics mainly measure output behavior and cannot reliably determine whether the target information is still stored internally or recoverable through alternative prompts or attacks; better evaluations of true removal versus hiding are needed. Second, SCALPEL modifies activations at inference and does not erase the parametric trace: the original information may persist at other layers or become accessible if the intervention is disabled, bypassed, or challenged. Our results therefore establish selective behavioral and representation-level unlearning *under the proposed intervention*, not certified deletion from the model. A natural next step is to use the extracted features to guide permanent parameter updates while preserving the selectivity of the representation-level approach; we view the present work as a first step toward bridging mechanistic interpretability and machine unlearning.